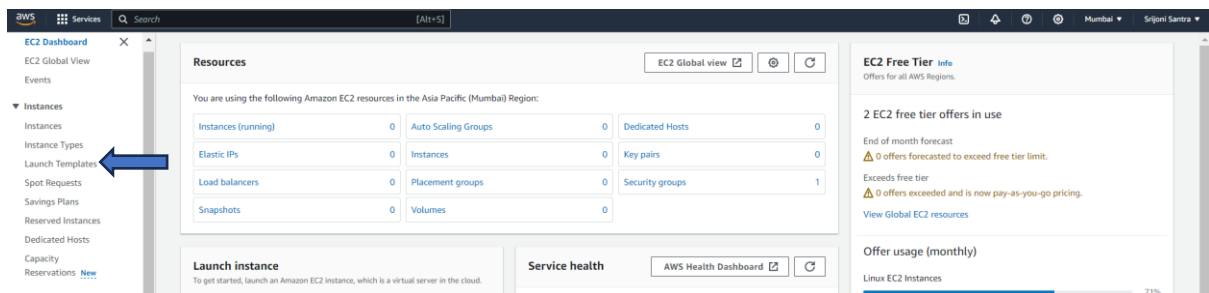


ASSIGNMENT – II

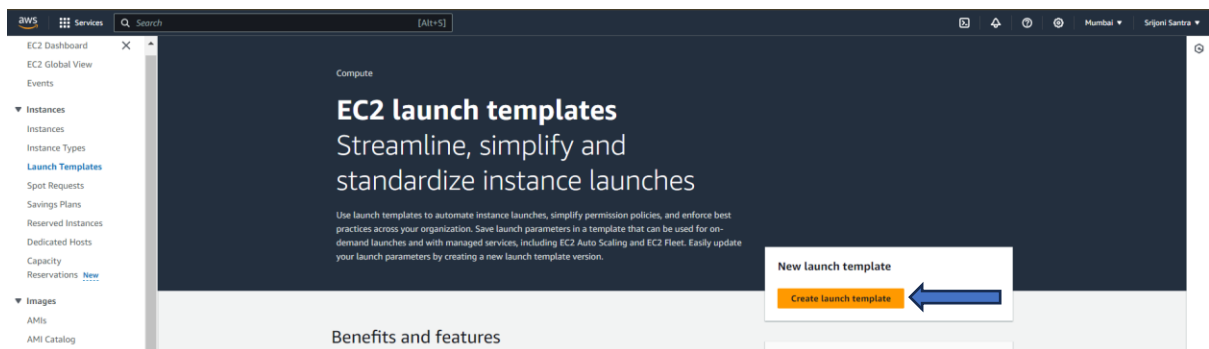
Problem Statement: Build a scaling plan in AWS that balance the load on different EC2 instances

Steps and Procedure

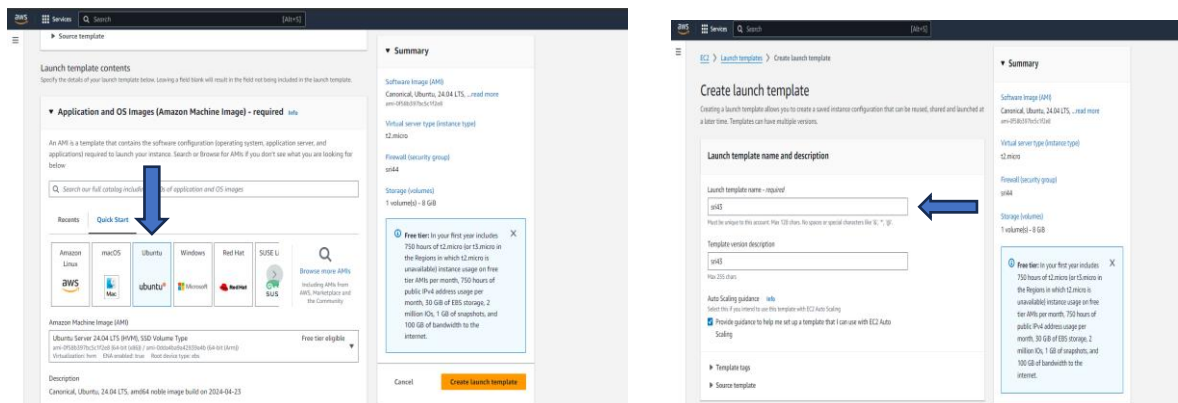
Step 1: Login to the console and click on EC2. Then click on “Launch Templates”.



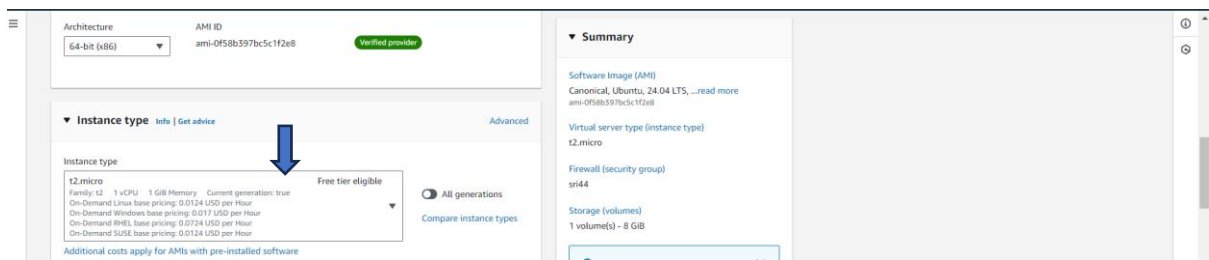
Step 2: Click on “Create launch template”.



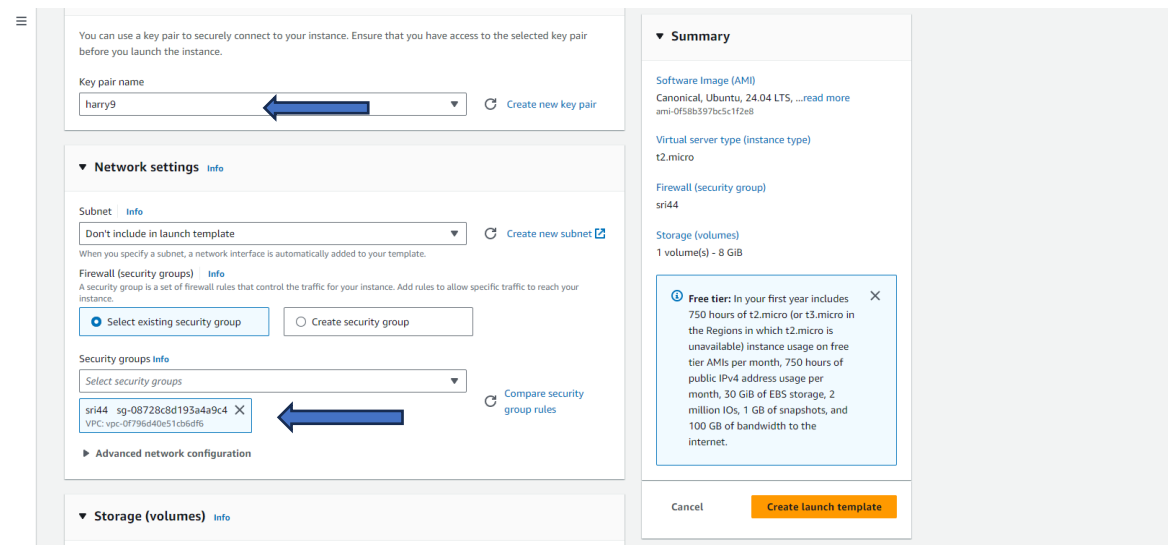
Step 3: Enter **Launch template name, description** and choose the option **Ubuntu** from **Quick start**.



Step 4: From **instance type** select the option **t2. Micro** or **t3. Micro**.



Step 5: Select the **key pair** and **security group**.



Step 6: The click on **advance details**, in **User data** write the commands and click on **create launch template**

```
#!/bin/bash
```

```
apt-get update
```

```
apt-get install -y nginx
```

```
systemctl start nginx
```

```
systemctl enable nginx
```

```
apt-get install -y git
```

```
curl -SL https://deb.nodesource.com/setup_16.x|sudo -E bash -
```

```
apt-get install -y nodejs
```

```
git clone http://github.com/sri2445/repo2.git
```

```
cd repo2
```

```
npm install
```

```
node index.js
```


Step 9: Enter the name of **auto scaling group**, select the **template name** and click on **Next**.

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console. The left sidebar lists steps from 1 to 7. The main content area is titled 'Choose launch template' and includes a description: 'Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.'

Under the 'Name' section, there is a text input field for the 'Auto Scaling group name' with the value 'hermione123'. A blue arrow points to this field.

Under the 'Launch template' section, there is a dropdown menu for the 'Launch template' with the value 'sri43'. A blue arrow points to this dropdown. Below the dropdown are links for 'Create a launch template' and 'Create a launch template version'.

At the bottom of the wizard, there is a 'Cancel' button and a 'Next' button. A blue arrow points to the 'Next' button.

Step 10: Select all the **availability zone** and click on **Next**.

The screenshot shows the 'Create Auto Scaling group' wizard in the AWS Management Console, specifically the 'Network' section. The left sidebar lists steps from 6 to 7. The main content area is titled 'Network' and includes a description: 'For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.'

Under the 'VPC' section, there is a dropdown menu for the 'VPC' with the value 'vpc-0f796d40e51cb6df6'. A blue arrow points to this dropdown.

Under the 'Availability Zones and subnets' section, there is a dropdown menu for 'Select Availability Zones and subnets' with the value 'ap-south-1a | subnet-0cec7545efdfa45d9'. A blue arrow points to this dropdown. Below the dropdown are three subnets listed: 'ap-south-1a | subnet-0cec7545efdfa45d9', 'ap-south-1b | subnet-0578f3b155eeb37c4', and 'ap-south-1c | subnet-0ea1190844a4b55ef'. A blue arrow points to the 'ap-south-1b' subnet.

At the bottom of the wizard, there is a 'Cancel' button, a 'Skip to review' button, a 'Previous' button, and a 'Next' button. A blue arrow points to the 'Next' button.

Step 11: In Configure advanced options select the option **attach to a new load balancer** & choose the load balancer scheme as **Internet facing**.

Step 11: In Configure advanced options select the option **attach to a new load balancer** & choose the load balancer scheme as **Internet facing**.

Step 12: Enter the **port number** & then from **Default routing** choose the option **Create a target group**.

Step 12: Enter the **port number** & then from **Default routing** choose the option **Create a target group**.

Step 13: Select the option **Turn on Elastic Load Balancing health checks** and you can change the health check grace period as per your requirement. Then click on Next.

The screenshot shows the 'Health checks' configuration page in the AWS Management Console. It includes sections for 'EC2 health checks' (Always enabled), 'Additional health check types - optional' (with 'Turn on Elastic Load Balancing health checks' checked and recommended), 'EC2 Auto Scaling will start to detect and act on health checks performed by Elastic Load Balancing' (with a warning to avoid unexpected terminations), 'Health check grace period' (set to 224 seconds), and 'Additional settings' (Monitoring and Default instance warmup). The 'Next' button is highlighted with a blue arrow.

Step 14: Enter the desired capacity, In desired capacity & Max desired capacity.

The screenshot shows the 'Configure group size and scaling' page in the AWS Management Console. It includes sections for 'Group size' (Desired capacity type: Units (number of instances), Desired capacity: 2) and 'Scaling' (Scaling limits: Min desired capacity: 2, Max desired capacity: 3). The 'Next' button is highlighted with a blue arrow.

Step 15: Choose the option **Target tracking scaling policy** and enter the **instance warmup** as per your requirement & then click on **Next**.

The screenshot shows the 'Scaling info' section of the AWS Auto Scaling console. It includes fields for 'Min desired capacity' (2) and 'Max desired capacity' (3). Under 'Automatic scaling - optional', the 'Target tracking scaling policy' is selected. The 'Scaling policy name' is 'Target Tracking Policy'. The 'Metric type' is 'Average CPU utilization' and the 'Target value' is 50. The 'Instance warmup' is set to 240 seconds. A 'Next' button is visible at the bottom right.

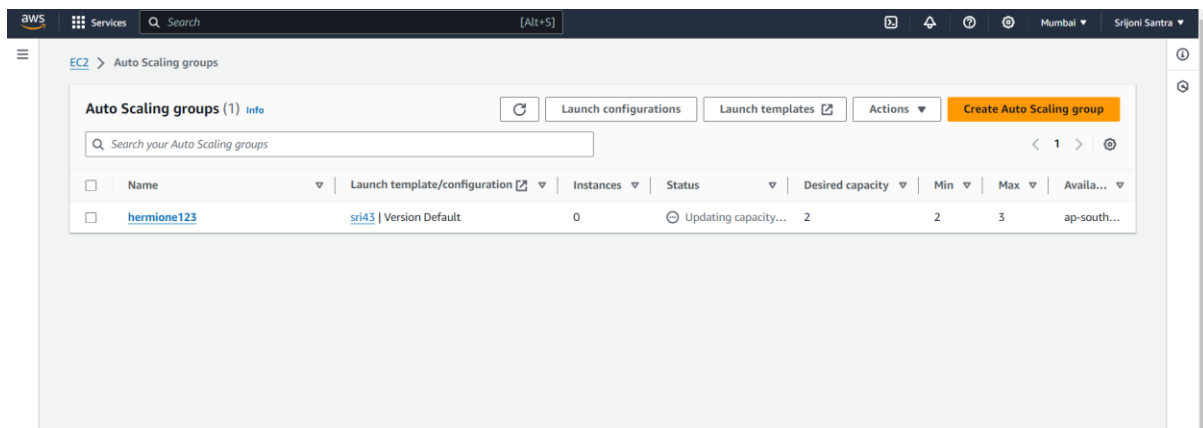
Step 16: Then click on **Next** Until the **review** page appears.

The screenshot shows the 'Add notifications - optional' page. It includes a text area for 'Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.' and an 'Add notification' button. At the bottom, there are 'Cancel', 'Skip to review', 'Previous', and 'Next' buttons.

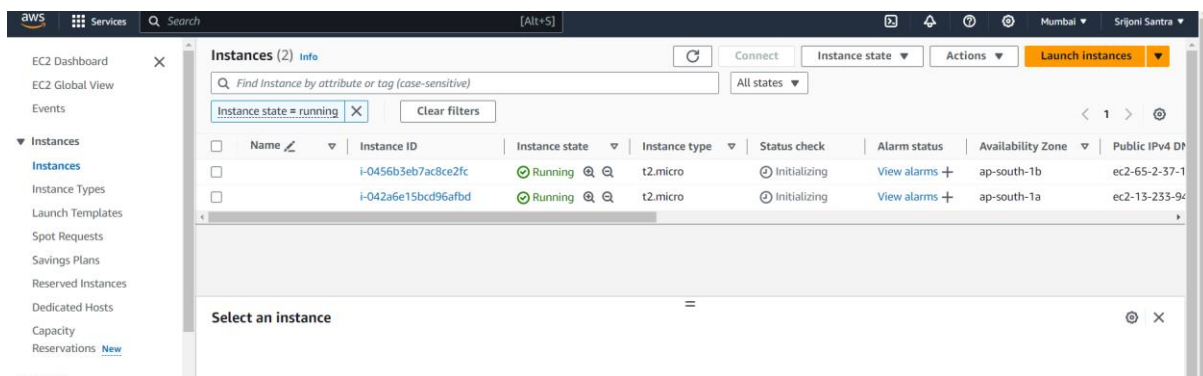
Step 17: Check all the review page and click on **Create auto scaling group**.

The screenshot shows the 'Review' page of the AWS Auto Scaling console. It displays a summary of the configuration, including 'Group details' (Auto Scaling group name: my-ec2-asg, Launch template: my-ec2-lt) and 'Step 2: Choose instance launch options' (Network: vpc-01234567890123456, Availability Zone: eu-west-1a, Subnet: subnet-01234567890123456). Below the summary, there are sections for 'Step 5: Add notifications' (No notifications) and 'Step 6: Add tags' (No tags). At the bottom, there are 'Cancel', 'Previous', and 'Create Auto Scaling group' buttons. A blue arrow points to the 'Create Auto Scaling group' button.

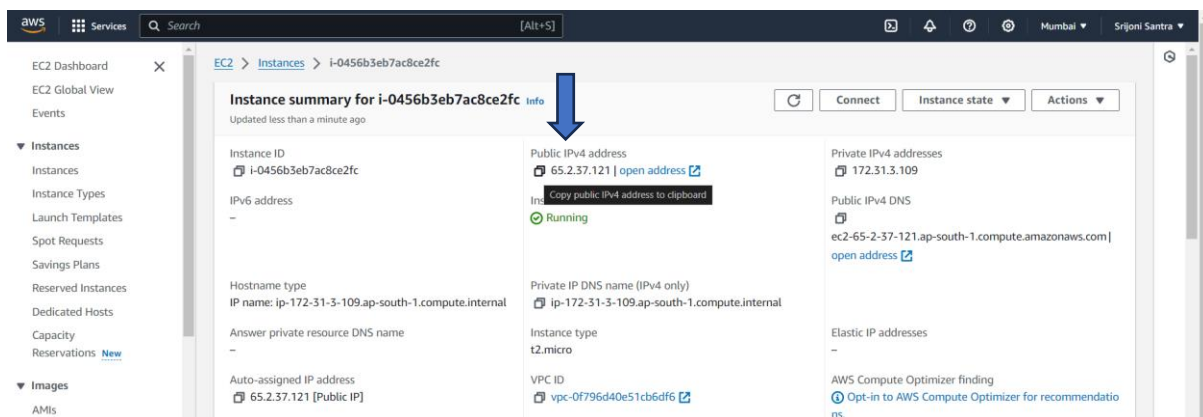
Step 18: The Auto scaling group is created successfully.



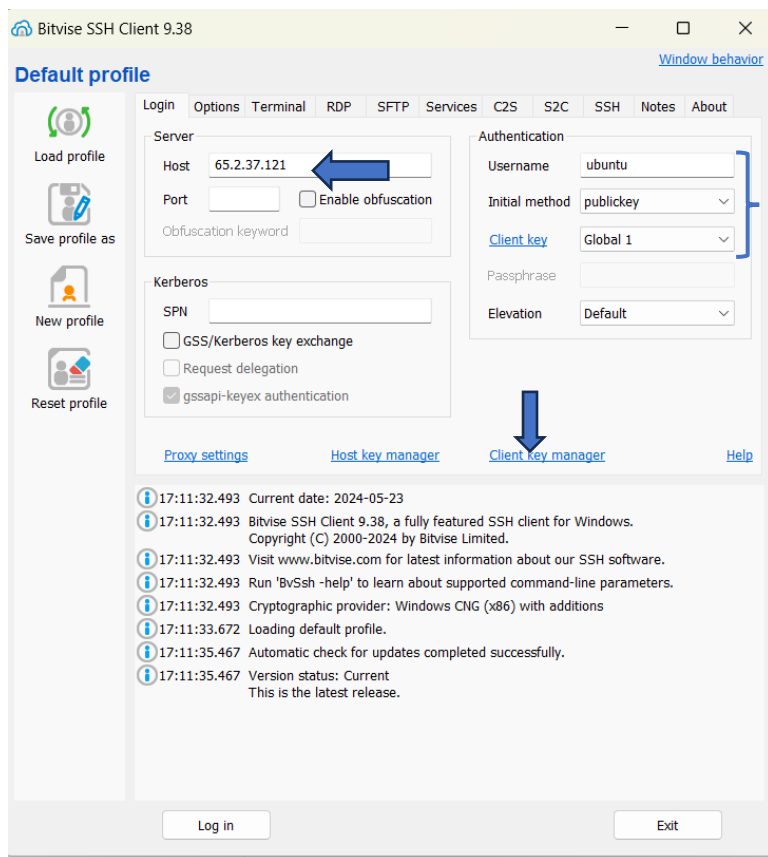
Step 19: Now return to the EC2 dashboard and check the instances running or not.



Step 20: Click on any instance id and copy the public IPv4 address.



Step 21: Paste the address under the host tab. Under the authentication tab, give the username as ubuntu, Initial method as publickey. Then click on Client key Manager.



Step 22: Remove any previously selected key if any, then click on import. Select the key using which instance was created. Then close the window & click on login. Now open a new terminal by clicking on New Terminal console. Write the commands properly in the terminal.

Command: `sudo nano infy.sh`

```

ubuntu@65.2.37.121:22 - Bitvise xterm - ubuntu@ip-172-31-3-109: ~
* Documentation: https://help.ubuntu.com
* Management:   https://landscape.canonical.com
* Support:      https://ubuntu.com/pro

System information as of Thu May 23 11:43:15 UTC 2024

System load:  0.12          Processes:      117
Usage of /:   29.3% of 6.71GB Users logged in:  0
Memory usage: 26%          IPv4 address for enX0: 172.31.3.109
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

23 updates can be applied immediately.
19 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

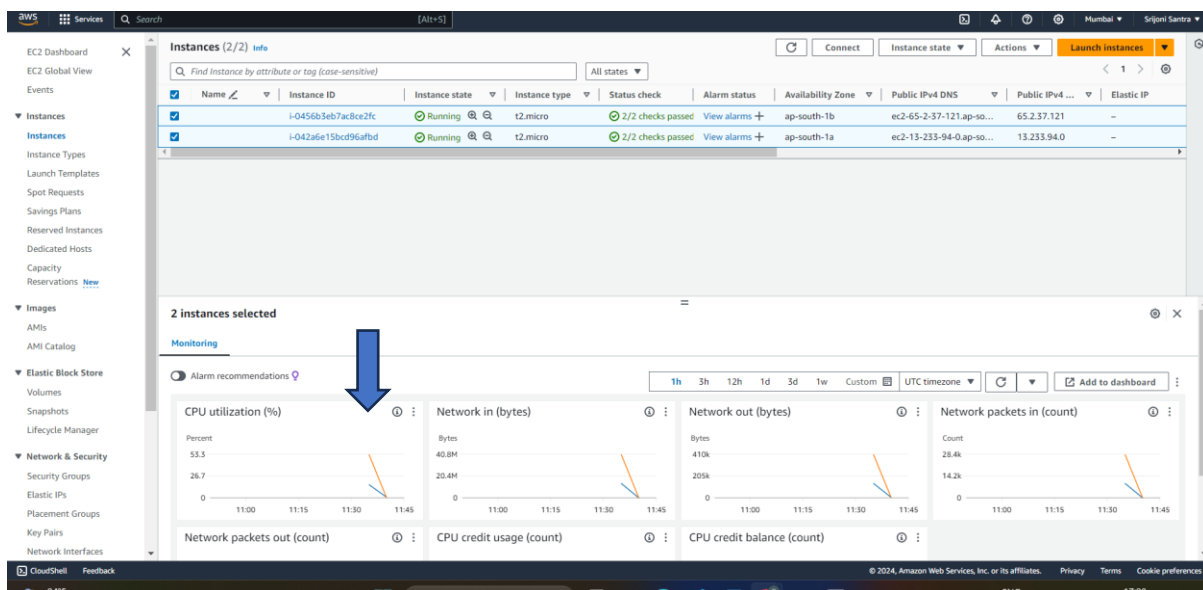
ubuntu@ip-172-31-3-109:~$ sudo nano infy.sh
ubuntu@ip-172-31-3-109:~$

```

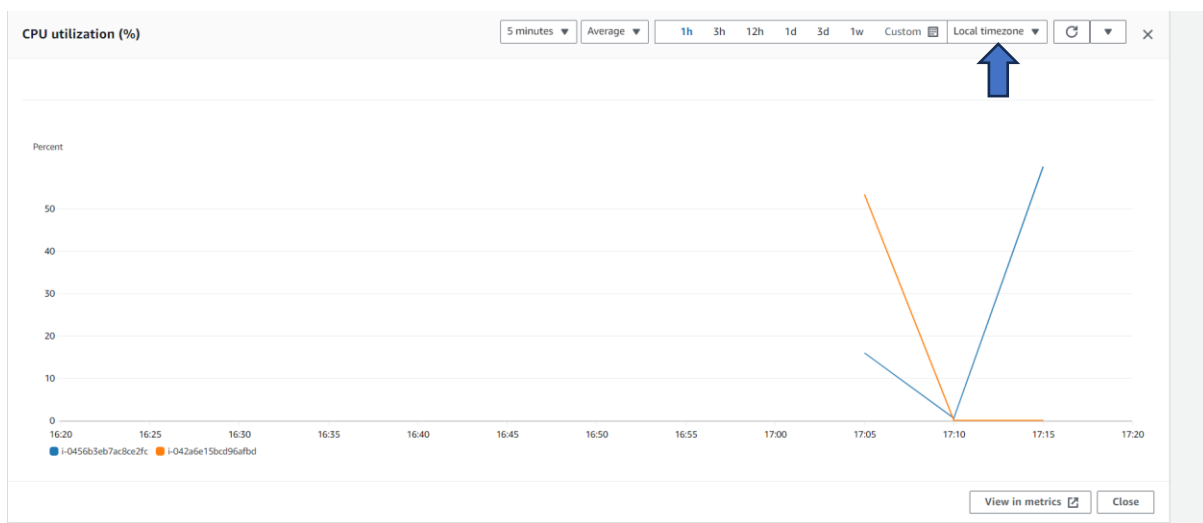
Step 23: Write the following shell program to run a infinity loop.

```
ubuntu@65.2.37.121:22 - Bitvise xterm - ubuntu@ip-172-31-3-109: ~
GNU nano 7.2 infy.sh *
#!/bin/bash
while(true)
do
echo "Inside loop"
done
```

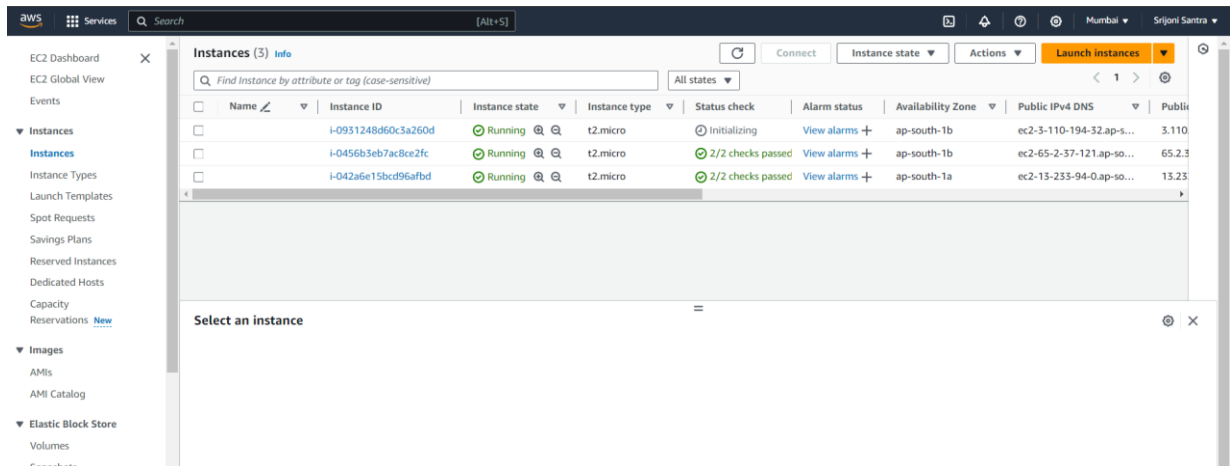
Step 24: Return to the instances and select the instances, then we can see the **CPU utilization**.



Step 25: Then enlarge it and change the timezone in **Local timezone**.

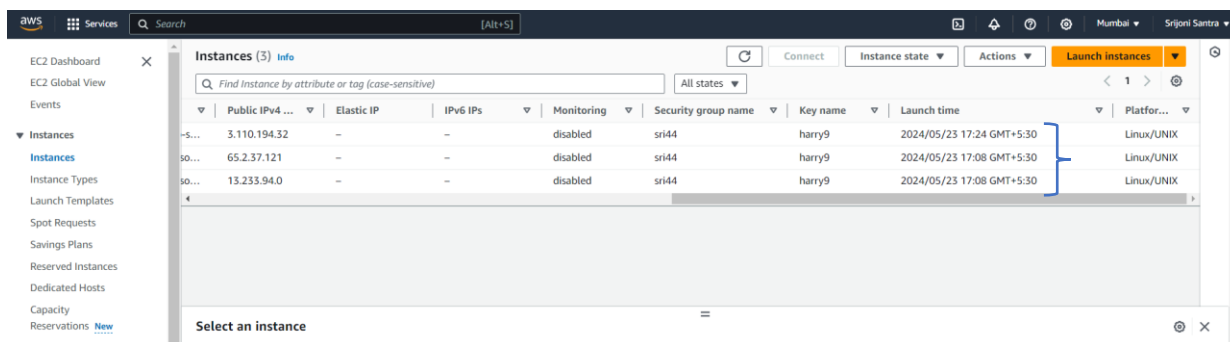


Step 26: Now if we check the instances, we can see a new instance is created after few minutes of the previous two instances.



The screenshot shows the AWS Management Console interface for the EC2 service. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity, and Reservations. The main content area displays the 'Instances (3)' page. A table lists three instances, all in a 'Running' state. The third instance, i-042a6e15bcd96afbd, is highlighted. Below the table, there is a 'Select an instance' prompt.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DNS	Public IP
	i-0931248d60c3a260d	Running	t2.micro	Initializing	View alarms +	ap-south-1b	ec2-3-110-194-32.ap-s...	3.110.
	i-0456b5eb7ac8ce2fc	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1b	ec2-65-2-37-121.ap-so...	65.2.3
	i-042a6e15bcd96afbd	Running	t2.micro	2/2 checks passed	View alarms +	ap-south-1a	ec2-13-233-94-0.ap-so...	13.23



The screenshot shows the AWS Management Console interface for the EC2 service. The left sidebar contains navigation links for EC2 Dashboard, EC2 Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, Savings Plans, Reserved Instances, Dedicated Hosts, Capacity, and Reservations. The main content area displays the 'Instances (3)' page. A table lists three instances, all in a 'Running' state. The third instance, i-042a6e15bcd96afbd, is highlighted. Below the table, there is a 'Select an instance' prompt.

Public IPv4 ...	Elastic IP	IPv6 IPs	Monitoring	Security group name	Key name	Launch time	Platform...
3.110.194.32	-	-	disabled	sri44	harry9	2024/05/23 17:24 GMT+5:30	Linux/UNIX
65.2.37.121	-	-	disabled	sri44	harry9	2024/05/23 17:08 GMT+5:30	Linux/UNIX
13.233.94.0	-	-	disabled	sri44	harry9	2024/05/23 17:08 GMT+5:30	Linux/UNIX